Converting Mass, Area and Volume

Lesson Plan

1. Lesson Information

Course	Grade 7 Accelerated (1205050)	Unit	5 — Multi-Step Proportional Relationships
Lesson	Lessons 8–9 — Converting Mass, Area and Volume	Date	_____
Length	53 minutes (one period)	Benchmark(s)	MA.7.AR.3.3

2. Standards

MA.7.AR.3.3 Solve mathematical and real-world problems involving the conversion of units across different measurement systems.

Must. Convert mass and volume with a printed reference-sheet ratio, either direction; convert area by squaring the length ratio (and volume by cubing it, in writing); route fluid ounces through cups. (MA.7.AR.3.3.)

Must not. A ratio not on the FAST reference sheet (no fl oz → mL, no ft^2 → m^2 as one number); a density or a weight-from-volume question; the capacity of a solid from its dimensions (Unit 6).

3. Mathematical Thinking and Reasoning Standards

the evidence for each, from this period

MTR	What it asks	The evidence in this lesson
MA.K12.MTR.5.1	Use patterns and structure to help understand and connect mathematical concepts.	Notes II: yesterday's k^2 and today's 144 are the same structure, and the room is asked to say so
MA.K12.MTR.4.1	Engage in discussions that reflect on the mathematical thinking of self and others.	boards 4 and 6 split; the sixty-second re-vote
MA.K12.MTR.6.1	Assess the reasonableness of solutions.	every mass and volume board says its estimate first — fewer kilograms than pounds, more liters than gallons

4. Learning Target and Essential Question

Learning target. I can convert mass, volume and area between systems, and square the length ratio for area.

Essential question. 1 ft = 12 in. Why is 1 ft^2 not 12 in^2?

Building on. 5.06–07 — multiply by the sheet's ratio, estimate first; 5.02–03 — area is two lengths, so k is used twice; Grade 6 — 1 ft^2 is a 12 by 12 square.

Working toward. 5.10–11 — a conversion inside a multi-step problem; Unit 6 — surface area and volume in whichever unit the problem names.

5. Sequence

read from the deck; never typed

Segment	Min	What the teacher does	What students do
Settle in; post the learning target	1	Post the target; say the one line the Teacher's Edition gives for it, and nothing else yet.	Copy the target.
Warm-Up	6	Two minutes silent, then reveal. One sentence of reteach per question at most. Then the new words, read once.	Four retrieval questions, alone, no notes. Then copy the new words.
Notes	10	Teach from the front; say the OFF line out loud.	Copy the notes; answer the checks aloud.
Example 1	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Example 2	4	Work the example; name the wrong boards before they appear.	Watch, then work the Your Turn alone.
Your Turn	3	Circulate; reveal and name what a wrong board most likely did.	Work it alone; boards up on the cue.
Whiteboards	11	One question at a time. Boards down until the cue, all up together; scan the back row first.	Answer on a board; hold it up on three.
Independent Set	6	Silent. Circulate and mark what you see; do not teach.	Six questions, written, alone.
IXL	5	Assign the day's IXL skills.	Start the skills; finish tonight.
Total	**53**		

6. Gradual Release

I do. The Notes slides and the worked half of each Example: the teacher writes, the class copies, and the OFF line is said out loud rather than printed.

We do. Each Example's Your Turn: the same steps on the student's own numbers, revealed and named a minute later.

You do. The nine-question whiteboard round (11 minutes) and then the six-question independent set, silent and written.

7. Higher-Order Questions

Question	DOK
1 m = 100 cm. Why is 1 m^2 = 10,000 cm^2 and not 100?	DOK 3
A 5-lb bag and a 2-kg bag: which is heavier, and what has to happen before you can say?	DOK 2
Why do fluid ounces go through cups to reach liters on this reference sheet?	DOK 2

8. Checks for Understanding

each board, and what to do about it

#	What the board asks	Answer	If the board is wrong
1	$1\,\text{kg} = 2.204\,\text{lb}$. A bag of rice is 10 kg. How many pounds is that?	***22.04 lb***	4.54 — × 0.454, the other ratio [B1G-M 7.AR.3.3: the inverted ratio]; 12.204 — added
2	$1\,\text{gal} = 3.785\,\text{L}$. A cooler holds 4 gal. How many liters is that?	***15.14 L***	1.06 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 7.785 — added
3	$1\,\text{L} = 0.264\,\text{gal}$. A tank holds 10 L. How many gallons is that?	***2.64 gal***	37.85 — × 3.785 [Notes I: pick the ratio whose 1 is the unit you have]; 0.264 — forgot the 10
4	$1\,\text{yd} = 3\,\text{ft}$. A patch of lawn is $2\,\text{yd}^2$ in area. How many square feet is that?	***B — 18 ft²***	(A) 2 × 6: 'two dimensions, so double the ratio' [5.02–03 Notes II: k × k, not 2k]; (C) 3^2 alone: the 2 yd^2 forgotten — one square yard, not two [4.06: finish the computation]; (D) 2 × 3: the ratio used once [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II]
5	$1\,\text{c} = 8\,\text{fl oz}$ and $1\,\text{c} = 0.24\,\text{L}$. A bottle holds 16 fl oz. How many liters is that?	***0.48 L***	3.84 — 16 × 0.24, the fluid ounces treated as cups [Notes III: the sheet's path is fl oz → c → L]; 453.6 — used 28.35, the weight ounce [vocabulary: oz and fl oz are two units]
6	$1\,\text{m} = 100\,\text{cm}$. How many square centimeters is $1\,\text{m}^2$?	***C — 10,000 cm²***	(A) 10 × 100: the ratio once, then a guess upward [7.AR.3.3: square units need the ratio squared — the unit's own]; (B) 2 × 100: 'two dimensions, so double it' [5.02–03 Notes II]; (D) the length ratio handed in for an area [7.AR.3.3: square units need the ratio squared — the unit's own; Notes II]
7	$1\,\text{kg} = 2.204\,\text{lb}$. A 5-lb bag of flour and a 2.5-kg bag of sugar. Which is heavier, and by how much?	***The sugar, by 0.51 lb***	'the flour, because 5 > 2.5' — the numbers compared, not the weights [MTR.6.1: units first]; 'the flour, by 2.5 lb' — 5 − 2.5 in two units
8	$1\,\text{gal} = 3.785\,\text{L}$. A 20-gal fish tank, bought for $45, is being filled. How many liters does it hold?	***75.7 L***	5.28 — × 0.264 [B1G-M 7.AR.3.3: the inverted ratio]; 170.33 — the price used
9	$1\,\text{ft} = 12\,\text{in}$. Priya says $2\,\text{ft}^3$ is $24\,\text{in}^3$, because $2 \times 12 = 24$. Name the error and fix it.	***She used the ratio once. A cubic foot is 12 × 12 × 12 = 1,728 in³, so 2 ft³ = 3,456 in³.***	288 in^3 — the ratio squared, today's rule applied to a cube [Notes III: three lengths, three times]; 'it should be 24 ft' — the unit changed instead of the number

MTR.4.1 — the one peer move. When a board splits — no option holding about two-thirds of the room — sixty seconds of "convince the person next to you", then re-vote, then reveal. The minute comes out of the whiteboard remainder.

9. Differentiation

ESE / 504. Three lines on the board all period: length — the ratio once; area — twice; volume — three times. Every board is answered by pointing at one line first. A 12 by 12 grid stays up beside them.

ELL. **Mass** and **weight** are one word here; **ounce** and **fluid ounce** are two different things — say "weight ounce" and "liquid ounce" aloud. **Square** means two lengths; **cubic** means three.

Enrichment. Board 9 finishes early for some: ask for 1 m^3 in cm^3 (1,000,000) and whether 1 L = 1,000 cm^3 makes sense from it (1 L = 1,000 mL; a liter is a 10-cm cube).

10. Closure, Vocabulary, Materials, Practice

Closure. Board 9 is written work and it is the exit evidence: the error named (Priya used the ratio once, and a cubic foot is three lengths), and the fix ($12 \times 12 \times 12 = 1{,}728$ in^3 per ft^3, so 2 ft^3 = 3,456 in^3). A board that writes 3,456 with no sentence about three lengths has named the number, not the reason.

Vocabulary. **mass and weight**; **ounce (oz)**; **fluid ounce (fl oz)**; **square units**; **cubic units**

Materials. Whiteboards and markers. The FAST reference sheet's conversion lines on the board all period (printed on every item as well); a 12 by 12 grid under the document camera for Notes II. The Unit Review paper to send home. Calculators are allowed on everything and printed on nothing.

Practice. The six-question independent set is worked silently in class; whatever is not finished goes home. IXL, all skills required to a SmartScore of 67, due by the Unit 5 test: Convert between customary and metric systems — E8Z; Compare and convert customary units — 4B8; Compare and convert metric units — HW2.